

Chapter 7: Limits

EXERCISE 7.1 [PAGE 100]

Exercise 7.1 | Q 1.1 | Page 100

Evaluate the following limits: $\lim_{x \rightarrow 3} \left[\frac{\sqrt{x+6}}{x} \right]$

SOLUTION

$$\begin{aligned} & \lim_{x \rightarrow 3} \left[\frac{\sqrt{x+6}}{x} \right] \\ &= \frac{\lim_{x \rightarrow 3} \sqrt{x+6}}{\lim_{x \rightarrow 3} x} \\ &= \frac{\sqrt{3+6}}{3} \\ &= \frac{\sqrt{9}}{3} \\ &= \frac{3}{3} \\ &= 1. \end{aligned}$$

Exercise 7.1 | Q 1.2 | Page 100

Evaluate the following limits: $\lim_{x \rightarrow 2} \left[\frac{x^{-3} - 2^{-3}}{x - 2} \right]$

SOLUTION

$$\begin{aligned} & \lim_{x \rightarrow 2} \left[\frac{x^{-3} - 2^{-3}}{x - 2} \right] \\ &= (-3) \cdot (2)^{-4} \quad \dots \left[\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right] \end{aligned}$$

$$= -3 \times \frac{1}{2^4}$$

$$= \frac{-3}{16}$$

Exercise 7.1 | Q 1.3 | Page 100

Evaluate the following limits: $\lim_{x \rightarrow 5} \left[\frac{x^3 - 125}{x^5 - 3125} \right]$

SOLUTION

$$\lim_{x \rightarrow 5} \left[\frac{x^3 - 125}{x^5 - 3125} \right]$$

$$= \lim_{x \rightarrow 5} \frac{\left(\frac{x^3 - 5^3}{x - 5} \right)}{\left(\frac{x^5 - 5^5}{x - 5} \right)} \quad \dots \left[\begin{array}{l} \because x \rightarrow 5 \therefore x \neq 5 \\ \therefore x - 5 \neq 0 \end{array} \right]$$

$$= \frac{\lim_{x \rightarrow 5} \frac{x^3 - 5^3}{x - 5}}{\lim_{x \rightarrow 5} \frac{x^5 - 5^5}{x - 5}}$$

$$= \frac{3(5)^2}{5(5)^4} \quad \dots \left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n \cdot a^{n-1} \right]$$

$$= \frac{3}{(5)^3}$$

$$= \frac{3}{125}.$$

Exercise 7.1 | Q 1.4 | Page 100

Evaluate the following limits: if $\lim_{x \rightarrow 1} \left[\frac{x^4 - 1}{x - 1} \right] = \lim_{x \rightarrow a} \left[\frac{x^3 - a^3}{x - a} \right]$, find all the value of a.

SOLUTION

$$\lim_{x \rightarrow 1} \left[\frac{x^4 - 1}{x - 1} \right] = \lim_{x \rightarrow a} \left[\frac{x^3 - a^3}{x - a} \right]$$

$$\lim_{x \rightarrow 1} \frac{x^4 - (1)^4}{x - 1} = \lim_{x \rightarrow a} \frac{x^3 - a^3}{x - a}$$

$$\therefore 4(1)^3 = 3a^2 \quad \dots \left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right]$$

$$\therefore 3a^2 = 4$$

$$\therefore a^2 = \frac{4}{3}$$

$$\therefore a = \pm \frac{2}{\sqrt{3}}.$$

Exercise 7.1 | Q 2.1 | Page 100

Evaluate the following limits: $\lim_{x \rightarrow 7} \left[\frac{(\sqrt[3]{x} - \sqrt[3]{7})(\sqrt[3]{x} + \sqrt[3]{7})}{x - 7} \right]$

SOLUTION

$$\lim_{x \rightarrow 7} \left[\frac{(\sqrt[3]{x} - \sqrt[3]{7})(\sqrt[3]{x} + \sqrt[3]{7})}{x - 7} \right]$$

$$= \lim_{x \rightarrow 7} \left[\frac{(x^{\frac{1}{3}} - 7^{\frac{1}{3}})(x^{\frac{1}{3}} + 7^{\frac{1}{3}})}{x - 7} \right]$$

$$= \lim_{x \rightarrow 7} \left[\frac{x^{\frac{2}{3}} 7^{\frac{2}{3}}}{x - 7} \right] \quad \dots [\because (a - b)(a + b) = a^2 - b^2]$$

$$= \frac{2}{7} (7)^{-\frac{1}{3}} \quad \dots \left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right]$$

$$= \frac{2}{3} \cdot \frac{1}{7^{\frac{1}{3}}}$$

$$= \frac{2}{3\sqrt[3]{7}}.$$

Exercise 7.1 | Q 2.2 | Page 100

Evaluate the following limits: if $\lim_{x \rightarrow 5} \left[\frac{x^k - 5^k}{x - 5} \right] = 500$, find all possible values of k .

SOLUTION

$$\lim_{x \rightarrow 5} \left[\frac{x^k - 5^k}{x - 5} \right] = 500$$

$$\therefore k(5)^{k-1} = 500 \quad \dots \left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right]$$

$$\therefore k(5)^{k-1} = 4 \times 125$$

$$\therefore k(5)^{k-1} = 4 \times (5)^3$$

$$\therefore k(5)^{k-1} = 4 \times (5)^{4-1}$$

Comparing both sides, we get

$$k = 4.$$

Exercise 7.1 | Q 2.3 | Page 100

Evaluate the following limits: $\lim_{x \rightarrow 0} \left[\frac{(1-x)^8 - 1}{(1-x)^2 - 1} \right]$

SOLUTION

$$\lim_{x \rightarrow 0} \frac{(1-x)^8 - 1}{(1-x)^2 - 1}$$

Put $1 - x = y$

As $x \rightarrow 0$, $y \rightarrow 1$

$$\begin{aligned}
&\therefore \lim_{x \rightarrow 0} \frac{(1-x)^8 - 1}{(1-x)^2 - 1} \\
&= \lim_{y \rightarrow 1} \frac{y^8 - 1^8}{y^2 - 1^2} \\
&= \lim_{y \rightarrow 1} \frac{\frac{y^8 - 1^8}{y-1}}{\frac{y^2 - 1^2}{y-1}} \quad \dots \left[\begin{array}{l} \because y \rightarrow 1 \therefore y \neq 1 \\ \therefore y - 1 \neq 0 \end{array} \right] \\
&= \frac{\lim_{y \rightarrow 1} \frac{y^8 - 1^8}{y-1}}{\lim_{y \rightarrow 1} \frac{y^2 - 1^2}{y-1}} \\
&= \frac{8(1)^7}{2(1)^1} \quad \dots \left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right] \\
&= 4
\end{aligned}$$

Alternative method:

$$\lim_{x \rightarrow 0} \frac{(1-x)^8 - 1}{(1-x)^2 - 1}$$

Put $1 - x = y$

As $x \rightarrow 0$, $y \rightarrow 1$

$$\begin{aligned}
&\therefore \lim_{x \rightarrow 0} \frac{(1-x)^8 - 1}{(1-x)^2 - 1} \\
&= \lim_{y \rightarrow 1} \frac{y^8 - 1}{y^2 - 1} \\
&= \lim_{y \rightarrow 1} \frac{(y^4 - 1)(y^4 + 1)}{y^2 - 1} \\
&= \lim_{y \rightarrow 1} \frac{(y^2 - 1)(y^2 + 1)(y^4 + 1)}{y^2 - 1}
\end{aligned}$$

$$= \lim_{y \rightarrow 1} (y^2 + 1)(y^4 + 1) \dots \left[\begin{array}{l} \because y \rightarrow 1 \therefore y \neq 1 \\ \therefore y^2 \neq 1 \\ \therefore y^2 - 1 \neq 0 \end{array} \right]$$

$$= (2)(2)$$

$$= 4.$$

Exercise 7.1 | Q 3.1 | Page 100

Evaluate the following limits: $\lim_{x \rightarrow 0} \left[\frac{\sqrt[3]{1+x} - \sqrt{1+x}}{x} \right]$

SOLUTION

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x} - \sqrt{1+x}}{x} \\ &= \lim_{x \rightarrow 0} \frac{(1+x)^{\frac{1}{3}} - (1+x)^{\frac{1}{2}}}{x} \end{aligned}$$

Put $1+x = y$

As $x \rightarrow 0, y \rightarrow 1$

$$\begin{aligned} & \therefore \lim_{x \rightarrow 0} \frac{(1+x)^{\frac{1}{3}} - (1+x)^{\frac{1}{2}}}{x} \\ &= \lim_{y \rightarrow 1} \frac{y^{\frac{1}{3}} - y^{\frac{1}{2}}}{y - 1} \\ &= \lim_{y \rightarrow 1} \frac{(y^{\frac{1}{3}} - 1) - (y^{\frac{1}{2}} - 1)}{y - 1} \\ &= \lim_{y \rightarrow 1} \left(\frac{y^{\frac{1}{3}} - 1}{y - 1} - \frac{y^{\frac{1}{2}} - 1}{y - 1} \right) \\ &= \lim_{y \rightarrow 1} \frac{y^{\frac{1}{3}} - 1^{\frac{1}{3}}}{y - 1} - \lim_{y \rightarrow 1} \frac{y^{\frac{1}{2}} - 1^{\frac{1}{2}}}{y - 1} \end{aligned}$$

$$\begin{aligned}
&= \lim_{y \rightarrow 1} \frac{y^{\frac{1}{3}} - 1^{\frac{1}{3}}}{y - 1} - \lim_{y \rightarrow 1} \frac{y^{\frac{1}{2}} - 1^{\frac{1}{2}}}{y - 1} \\
&= \frac{1}{3} (1)^{\frac{-2}{3}} - \frac{1}{2} (1)^{\frac{-1}{2}} \quad \dots \left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right] \\
&= \frac{1}{3} - \frac{1}{2} \\
&= \frac{2 - 3}{6} \\
&= -\frac{1}{6}.
\end{aligned}$$

Exercise 7.1 | Q 3.2 | Page 100

Evaluate the following limits: $\lim_{y \rightarrow 1} \left[\frac{2y - 2}{\sqrt[3]{7+y} - 2} \right]$

SOLUTION

$$\begin{aligned}
&\lim_{y \rightarrow 1} \frac{2y - 2}{\sqrt[3]{7+y} - 2} \\
&= \lim_{y \rightarrow 1} \frac{2(y - 1)}{(7 + y)^{\frac{1}{3}} - 8^{\frac{1}{3}}} \quad \dots \left[\because 2 = (2^3)^{\frac{1}{3}} = 8^{\frac{1}{3}} \right] \\
&= \lim_{y \rightarrow 1} \frac{2}{\frac{(y+7)^{\frac{1}{3}} - 8^{\frac{1}{3}}}{y-1}} \\
&= \frac{\lim_{y \rightarrow 1} 2}{\lim_{y \rightarrow 1} \frac{(y+7)^{\frac{1}{3}} - 8^{\frac{1}{3}}}{(y+7) - 8}}
\end{aligned}$$

Let $y + 7 = x$

As $y \rightarrow 1, x \rightarrow 8$

$$\begin{aligned}
&= \frac{2}{\lim_{x \rightarrow 8} \frac{x^{\frac{1}{3}} - 8^{\frac{1}{3}}}{x - 8}} \\
&= \frac{2}{\frac{1}{3}(8)^{-\frac{2}{3}}} \dots \left[\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right] \\
&= 2(3) \cdot (8)^{\frac{2}{3}} \\
&= 6(2^3)^{\frac{2}{3}} \\
&= 6 \times (2)^2 \\
&= 24.
\end{aligned}$$

Exercise 7.1 | Q 3.3 | Page 100

Evaluate the following limits: $\lim_{x \rightarrow a} \left[\frac{(z+2)^{\frac{3}{2}} - (a+2)^{\frac{3}{2}}}{z-a} \right]$

SOLUTION

$$\lim_{x \rightarrow a} \frac{(z+2)^{\frac{3}{2}} - (a+2)^{\frac{3}{2}}}{z-a}$$

Put $z+2 = y$ and $a+2 = b$

As $z \rightarrow a$, $z+2 \rightarrow a+2$

i.e. $y \rightarrow b$

$$\begin{aligned}
&\therefore \lim_{z \rightarrow a} \frac{(z+2)^{\frac{3}{2}} - (a+2)^{\frac{3}{2}}}{z-a} \\
&= \lim_{y \rightarrow b} \frac{y^{\frac{3}{2}} - b^{\frac{3}{2}}}{(y-2) - (b-2)} \\
&= \lim_{y \rightarrow b} \frac{y^{\frac{3}{2}} - b^{\frac{3}{2}}}{y-b}
\end{aligned}$$

$$= \frac{3}{2} \cdot b^{\frac{1}{2}} \quad \dots \left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right]$$

$$= \frac{3}{2} (a + 2)^{\frac{1}{2}} \quad \dots [\because b = a + 2]$$

Exercise 7.1 | Q 3.4 | Page 100

Evaluate the following limits: $\lim_{x \rightarrow 5} \left[\frac{x^3 - 125}{x^2 - 25} \right]$

SOLUTION

$$\lim_{x \rightarrow 5} \frac{x^3 - 125}{x^2 - 25}$$

$$= \lim_{x \rightarrow 5} \frac{\frac{x^3 - 125}{x^2 - 25}}{\frac{x^2 - 25}{x - 5}} \quad \dots \left[\begin{array}{l} \text{As } x \rightarrow 5, x \neq 5 \\ \therefore x - 5 \neq 0 \\ \text{Divide Numerator and} \\ \text{Denominator by } x - 5. \end{array} \right]$$

$$= \lim_{x \rightarrow 5} \frac{\left(\frac{x^3 - 5^3}{x - 5} \right)}{\left(\frac{x^2 - 5^2}{x - 5} \right)}$$

$$= \frac{3(5)^2}{2(5)^1} \quad \dots \left[\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right]$$

$$= \frac{15}{2}.$$

EXERCISE 7.2 [PAGE 102]

Exercise 7.2 | Q 1.1 | Page 102

Evaluate the following limits: $\lim_{x \rightarrow 2} \left[\frac{z^2 - 5z + 6}{z^2 - 4} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 2} \frac{z^2 - 5z + 6}{z^2 - 4} \\
&= \lim_{z \rightarrow 2} \frac{(z - 3)(z - 2)}{(z + 2)(z - 2)} \\
&= \lim_{x \rightarrow 2} \frac{z - 3}{z - 2} \quad \dots \left[\begin{array}{l} \text{As } z \rightarrow 2 \text{ } z \neq 2 \\ \therefore z - 2 \neq 0 \end{array} \right] \\
&= \frac{2 - 3}{2 + 2} \\
&= -\frac{1}{4}.
\end{aligned}$$

Exercise 7.2 | Q 1.2 | Page 102

Evaluate the following limits: $\lim_{x \rightarrow -3} \left[\frac{x + 3}{x^2 + 4x + 3} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow -3} \left[\frac{x + 3}{x^2 + 4x + 3} \right] \\
&= \lim_{x \rightarrow -3} \frac{x + 3}{(x + 3)(x + 1)} \\
&= \lim_{x \rightarrow -3} \frac{1}{x + 1} \quad \dots \left[\begin{array}{l} \text{As } x \rightarrow -3 \text{ } x \neq -3 \\ \therefore x + 3 \neq 0 \end{array} \right] \\
&= \frac{1}{-3 + 1} \\
&= -\frac{1}{2}.
\end{aligned}$$

Exercise 7.2 | Q 1.3 | Page 102

Evaluate the following limits: $\lim_{y \rightarrow 0} \left[\frac{5y^3 + 8y^2}{3y^4 - 16y^2} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{y \rightarrow 0} \left[\frac{5y^3 + 8y^2}{3y^4 - 16y^2} \right] \\
&= \lim_{y \rightarrow 0} \frac{y^2(5y + 8)}{y^2(3y^2 - 16)} \\
&= \lim_{y \rightarrow 0} \frac{5y + 8}{3y^2 - 16} \quad \cdots \left[\begin{array}{l} \text{As } y \rightarrow 0 \quad y \neq 0 \\ \therefore y^2 \neq 0 \end{array} \right] \\
&= \frac{5(0) + 8}{3(0)^2 - 16} \\
&= \frac{8}{-16} \\
&= -\frac{1}{2}.
\end{aligned}$$

Exercise 7.2 | Q 1.4 | Page 102

Evaluate the following limits: $\lim_{x \rightarrow -2} \left[\frac{-2x - 4}{x^3 + 2x^2} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow -2} \left[\frac{-2x - 4}{x^3 + 2x^2} \right] \\
&= \lim_{x \rightarrow -2} \frac{-2(x + 2)}{x^2(x + 2)} \\
&= \lim_{x \rightarrow -2} \frac{-2}{x^2} \quad \cdots \left[\begin{array}{l} \text{As } x \rightarrow -2 \quad x \neq -2 \\ \therefore x + 2 \neq 0 \end{array} \right] \\
&= \frac{(-2)}{(-2)^2} \\
&= \frac{-2}{4} \\
&= \frac{-1}{2}.
\end{aligned}$$

Evaluate the following limits: $\lim_{u \rightarrow 1} \left[\frac{u^4 - 1}{u^3 - 1} \right]$

SOLUTION

$$\begin{aligned}
 & \lim_{u \rightarrow 1} \left[\frac{u^4 - 1}{u^3 - 1} \right] \\
 &= \lim_{u \rightarrow 1} \left[\frac{\left[\frac{u^4 - 1^4}{u - 1} \right]}{\left[\frac{u^3 - 1^3}{u - 1} \right]} \right] \quad \dots \left[\begin{array}{l} \because u \rightarrow 1; u \neq 1 \\ \therefore u - 1 \neq 0 \end{array} \right] \\
 &= \frac{4(1)^2}{3(1)^2} \quad \dots \left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right] \\
 &= \frac{4}{3}.
 \end{aligned}$$

Evaluate the following limits: $\lim_{x \rightarrow 3} \left[\frac{1}{x - 3} - \frac{9x}{x^3 - 27} \right]$

SOLUTION

$$\begin{aligned}
 & \lim_{x \rightarrow 3} \left[\frac{1}{x - 3} - \frac{9x}{x^3 - 27} \right] \\
 &= \lim_{x \rightarrow 3} \left[\frac{1}{x - 3} - \frac{9x}{x^3 - 3^3} \right] \\
 &= \lim_{x \rightarrow 3} \left[\frac{1}{x - 3} - \frac{9x}{x^3 - 3^3} \right] \\
 &= \lim_{x \rightarrow 3} \left[\frac{1}{x - 3} - \frac{9x}{(x - 3)(x^2 + 3x + 9)} \right] \\
 &= \lim_{x \rightarrow 3} \left[\frac{x^2 + 3x + 9 - 9x}{(x - 3)(x^2 + 3x + 9)} \right]
 \end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 3} \left[\frac{x^2 + 3x + 9 - 9x}{(x - 3)(x^2 + 3x + 9)} \right] \\
&= \lim_{x \rightarrow 3} \left[\frac{x^2 - 6x + 9}{(x - 3)(x^2 + 3x + 9)} \right] \\
&= \lim_{x \rightarrow 3} \left[\frac{(x - 3)^2}{(x - 3)(x^2 + 3x + 9)} \right] \\
&= \lim_{x \rightarrow 3} \left[\frac{x - 3}{x^2 + 3x + 9} \right] \quad \cdots \left[\begin{array}{l} \because x \rightarrow 3, x \neq 3 \\ \therefore x - 3 \neq 0 \end{array} \right] \\
&= \frac{3 - 3}{(3)^2 + 3(3) + 9} \\
&= \frac{0}{27} \\
&= 0.
\end{aligned}$$

Exercise 7.2 | Q 2.3 | Page 102

Evaluate the following limits: $\lim_{x \rightarrow 2} \left[\frac{x^3 - 4x^2 + 4x}{x^2 - 1} \right]$

SOLUTION

$$\begin{aligned}
&\lim_{x \rightarrow 2} \left[\frac{x^3 - 4x^2 + 4x}{x^2 - 1} \right] \\
&= \lim_{x \rightarrow 2} \frac{x(x^2 - 4x + 4)}{(x^2 - 1)} \\
&= \lim_{x \rightarrow 2} \frac{x(x - 2)^2}{x^2 - 1} \\
&= \frac{2(0)}{(2)^2 - 1}
\end{aligned}$$

$$= \frac{2 \times 0}{3}$$

$$= 0.$$

Exercise 7.2 | Q 3.1 | Page 102

Evaluate the following limits: $\lim_{x \rightarrow -2} \left[\frac{x^7 + x^5 + 160}{x^3 + 8} \right]$

SOLUTION

$$\lim_{x \rightarrow -2} \left[\frac{x^7 + x^5 + 160}{x^3 + 8} \right]$$

$$= \lim_{x \rightarrow -2} \frac{(x^7 + 128) + (x^5 + 32)}{x^3 + 8}$$

$$= \lim_{x \rightarrow -2} \frac{\frac{(x^7+128)+(x^5+32)}{x+2}}{\frac{x^3+8}{x+2}} \quad \dots \left[\begin{array}{l} \text{As } x \rightarrow -2, x \neq -2 \\ \therefore x + 2 \neq 0 \\ \text{Divide Numerator and} \\ \text{Denominator by } x + 2 \end{array} \right]$$

$$= \frac{\lim_{x \rightarrow -2} \left(\frac{x^7+2^7}{x+2} + \frac{x^5+2^5}{x+2} \right)}{\lim_{x \rightarrow -2} \left(\frac{x^3+2^3}{x+2} \right)}$$

$$= \frac{\lim_{x \rightarrow -2} \frac{x^7 - (-2)^7}{x - (-2)} + \lim_{x \rightarrow -2} \frac{x^5 - (-2)^5}{x - (-2)}}{\lim_{x \rightarrow -2} \frac{x^3 - (-2)^3}{x - (-2)}}$$

$$\begin{aligned}
&= \frac{7(-2)^6 + 5(-2)^4}{3(-2)^2} \dots \left[\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right] \\
&= \frac{7(64) + 5(16)}{3(4)} \\
&= \frac{448 + 80}{12} \\
&= \frac{528}{12} \\
&= 44.
\end{aligned}$$

Exercise 7.2 | Q 3.2 | Page 102

Evaluate the following limits: $\lim_{y \rightarrow \frac{1}{2}} \left[\frac{1 - 8y^3}{y - 4y^3} \right]$

SOLUTION

$$\begin{aligned}
&\lim_{y \rightarrow \frac{1}{2}} \left[\frac{1 - 8y^3}{y - 4y^3} \right] \\
&= \lim_{y \rightarrow \frac{1}{2}} \frac{1 - 8y^3}{y(1 - 4y^2)} \\
&= \lim_{y \rightarrow \frac{1}{2}} \frac{(1)^3 - (2y)^3}{y[(1)^2 - (2y)^2]} \\
&= \lim_{y \rightarrow \frac{1}{2}} \frac{(1 - 2y)(1 + 2y + 4y^2)}{y(1 - 2y)(1 + 2y)} \\
&= \lim_{y \rightarrow \frac{1}{2}} \frac{1 + 2y + 4y^2}{y(1 + 2y)} \dots \left[\begin{array}{l} \because y \rightarrow \frac{1}{2}, \therefore y \neq \frac{1}{2} \\ \therefore 2y \neq 1, \therefore 2y - 1 \neq 0 \\ \therefore 1 - 2y \neq 0 \end{array} \right] \\
&= \frac{1 + 2\left(\frac{1}{2}\right) + 4\left(\frac{1}{2}\right)^2}{\frac{1}{2} \left[1 + 2\left(\frac{1}{2}\right) \right]}
\end{aligned}$$

$$= \frac{1 + 1 + 1}{\frac{1}{2}(2)}$$

$$= 3.$$

Exercise 7.2 | Q 3.3 | Page 102

Evaluate the following limits: $\lim_{v \rightarrow \sqrt{2}} \left[\frac{v^2 + v\sqrt{2} - 4}{v^2 - 3v\sqrt{2} + 4} \right]$

SOLUTION

$$\lim_{v \rightarrow \sqrt{2}} \left[\frac{v^2 + v\sqrt{2} - 4}{v^2 - 3v\sqrt{2} + 4} \right]$$

$$\text{Consider, } v^2 + v\sqrt{2} - 4 = v^2 + \sqrt{2}v - 4$$

$$= v^2 + 2\sqrt{2}v - \sqrt{2}v - 4$$

$$= v(v + 2\sqrt{2}) - \sqrt{2}(v + 2\sqrt{2})$$

$$= (v + 2\sqrt{2})(v - \sqrt{2})$$

$$v^2 - 3v\sqrt{2} + 4 = v^2 - 3\sqrt{2}v + 4$$

$$= v^2 - 2\sqrt{2}v - \sqrt{2}v + 4$$

$$= v(v - 2\sqrt{2}) - \sqrt{2}(v - 2\sqrt{2})$$

$$= (v - 2\sqrt{2})(v - \sqrt{2})$$

$$\therefore \lim_{v \rightarrow \sqrt{2}} \left[\frac{v^2 + v\sqrt{2} - 4}{v^2 - 3v\sqrt{2} + 4} \right]$$

$$\begin{aligned}
&= \lim_{v \rightarrow \sqrt{2}} \frac{(v + 2\sqrt{2})(v - \sqrt{2})}{(v - 2\sqrt{2})(v - \sqrt{2})} \\
&= \lim_{v \rightarrow \sqrt{2}} \frac{v + 2\sqrt{2}}{v - 2\sqrt{2}} \quad \dots \left[\begin{array}{l} \text{As } v \rightarrow \sqrt{2}, v \neq \sqrt{2} \\ \therefore v - \sqrt{2} \neq 0 \end{array} \right] \\
&= \frac{\sqrt{2} + 2\sqrt{2}}{\sqrt{2} - 2\sqrt{2}} \\
&= \frac{3\sqrt{2}}{-\sqrt{2}} \\
&= -3.
\end{aligned}$$

Exercise 7.2 | Q 3.4 | Page 102

Evaluate the following limits: $\lim_{x \rightarrow 3} \left[\frac{x^2 + 2x - 15}{x^2 - 5x + 6} \right]$

SOLUTION

$$\begin{aligned}
&\lim_{x \rightarrow 3} \left[\frac{x^2 + 2x - 15}{x^2 - 5x + 6} \right] \\
&= \lim_{x \rightarrow 3} \frac{(x + 5)(x - 3)}{(x - 2)(x - 3)} \\
&= \lim_{x \rightarrow 3} \frac{x + 5}{x - 2} \quad \dots \left[\begin{array}{l} \text{as } x \rightarrow 3, x \neq 3 \\ \therefore x - 3 \neq 0 \end{array} \right] \\
&= \frac{3 + 5}{3 - 2} \\
&= 8.
\end{aligned}$$

EXERCISE 7.3 [PAGE 103]

Exercise 7.3 | Q 1.1 | Page 103

Evaluate the following limits: $\lim_{x \rightarrow 0} \left[\frac{\sqrt{6+x+x^2} - \sqrt{6}}{x} \right]$

SOLUTION

$$\begin{aligned}
 & \lim_{x \rightarrow 0} \left[\frac{\sqrt{6+x+x^2} - \sqrt{6}}{x} \right] \\
 &= \lim_{x \rightarrow 0} \left[\frac{\sqrt{6+x+x^2} - \sqrt{6}}{x} \times \frac{\sqrt{6+x+x^2} + \sqrt{6}}{\sqrt{6+x+x^2} + \sqrt{6}} \right] \\
 &= \lim_{x \rightarrow 0} \frac{(6+x+x^2) - 6}{x(\sqrt{6+x+x^2} + \sqrt{6})} \\
 &= \lim_{x \rightarrow 0} \frac{x+x^2}{x(\sqrt{6+x+x^2} + \sqrt{6})} \\
 &= \lim_{x \rightarrow 0} \frac{x(1+x)}{x(\sqrt{6+x+x^2} + \sqrt{6})} \\
 &= \lim_{x \rightarrow 0} \frac{1+x}{\sqrt{6+x+x^2} + \sqrt{6}} \quad \dots [\because x \rightarrow 0, \therefore x \neq 0] \\
 &= \frac{(1+0)}{\sqrt{6} + \sqrt{6}} \\
 &= \frac{1}{2\sqrt{6}}.
 \end{aligned}$$

Exercise 7.3 | Q 1.2 | Page 103

Evaluate the following limits: $\lim_{y \rightarrow 0} \left[\frac{\sqrt{1-y^2} - \sqrt{1+y^2}}{y^2} \right]$

SOLUTION

$$\begin{aligned}& \lim_{y \rightarrow 0} \left[\frac{\sqrt{1-y^2} - \sqrt{1+y^2}}{y^2} \right] \\&= \lim_{y \rightarrow 0} \left[\frac{\sqrt{1-y^2} - \sqrt{1+y^2}}{y^2} \right] \\&= \lim_{y \rightarrow 0} \left[\frac{\sqrt{1-y^2} - \sqrt{1+y^2}}{y^2} \times \frac{\sqrt{1-y^2} + \sqrt{1+y^2}}{\sqrt{1-y^2} + \sqrt{1+y^2}} \right] \\&= \lim_{y \rightarrow 0} \frac{(1-y^2) - (1+y^2)}{(\sqrt{1-y^2} + \sqrt{1+y^2})} \\&= \lim_{y \rightarrow 0} \frac{1-y^2-1-y^2}{y^2(\sqrt{1-y^2} + \sqrt{1+y^2})} \\&= \lim_{y \rightarrow 0} \frac{-2y^2}{y^2(\sqrt{1-y^2} + \sqrt{1+y^2})} \\&= \lim_{y \rightarrow 0} \frac{-2}{\sqrt{1-y^2} + \sqrt{1+y^2}} \cdots \left[\begin{array}{l} \because y \rightarrow 0, \therefore y \neq 0, \\ \therefore y^2 \neq 0 \end{array} \right] \\&= \frac{-2}{\sqrt{1-0^2} + \sqrt{1+0^2}} \\&= \frac{-2}{1+1} \\&= -1.\end{aligned}$$

Exercise 7.3 | Q 1.3 | Page 103

Evaluate the following limits: $\lim_{x \rightarrow 2} \left[\frac{\sqrt{2+x} - \sqrt{6-x}}{\sqrt{x} - \sqrt{2}} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 2} \left[\frac{\sqrt{2+x} - \sqrt{6-x}}{\sqrt{x} - \sqrt{2}} \right] \\
&= \lim_{x \rightarrow 2} \left[\frac{\sqrt{2+x} - \sqrt{6-x}}{\sqrt{x} - \sqrt{2}} \times \frac{\sqrt{2+x} + \sqrt{6-x}}{\sqrt{2+x} + \sqrt{6-x}} \times \frac{\sqrt{x} + \sqrt{2}}{\sqrt{x} + \sqrt{2}} \right] \dots \left[\begin{array}{l} \text{By taking conjugates} \\ \text{of both, the numerator} \\ \text{as well as the} \\ \text{Denominator} \end{array} \right] \\
&= \lim_{x \rightarrow 2} \left[\frac{(2+x) - (6-x)}{(x-2)} \times \frac{\sqrt{x} + \sqrt{2}}{\sqrt{2+x} + \sqrt{6-x}} \right] \\
&= \lim_{x \rightarrow 2} \left[\frac{-4+2x}{x-2} \times \frac{\sqrt{x} + \sqrt{2}}{\sqrt{2+x} + \sqrt{6-x}} \right] \\
&= \lim_{x \rightarrow 2} \left[\frac{2(x-2)}{x-2} \times \frac{\sqrt{x} + \sqrt{2}}{\sqrt{2+x} + \sqrt{6-x}} \right] \\
&= \lim_{x \rightarrow 2} \left[\frac{2(\sqrt{x} + \sqrt{2})}{\sqrt{2+x} + \sqrt{6-x}} \right] \dots \left[\begin{array}{l} \because x \rightarrow 2; x \neq 2 \\ \therefore x-2 \neq 0 \end{array} \right] \\
&= \frac{2(\sqrt{2} + \sqrt{2})}{\sqrt{2+2} + \sqrt{6-2}} \\
&= \frac{2(2\sqrt{2})}{2+2} \\
&= \frac{4\sqrt{2}}{4} \\
&= \sqrt{2}.
\end{aligned}$$

Exercise 7.3 | Q 2.1 | Page 103

Evaluate the following limits: $\lim_{x \rightarrow a} \left[\frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}} \right]$

SOLUTION

$$\begin{aligned}& \lim_{x \rightarrow a} \left[\frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}} \right] \\&= \lim_{x \rightarrow a} \left[\frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}} \times \frac{\sqrt{a+2x} + \sqrt{3x}}{\sqrt{a+2x} + \sqrt{3x}} \times \frac{\sqrt{3a+x} + 2\sqrt{x}}{\sqrt{3a+x} + 2\sqrt{x}} \right] \\&= \lim_{x \rightarrow a} \left[\frac{(a+2x) - 3x}{(3a+x) - 4x} \times \frac{\sqrt{3a+x} + 2\sqrt{x}}{\sqrt{a+2x} + \sqrt{3x}} \right] \\&= \lim_{x \rightarrow a} \left[\frac{a-x}{3a-3x} \times \frac{\sqrt{3a+x} + 2\sqrt{x}}{\sqrt{a+2x} + \sqrt{3x}} \right] \\&= \lim_{x \rightarrow a} \left[\frac{-(x-a)}{-3(x-a)} \times \frac{\sqrt{3a+x} + 2\sqrt{x}}{\sqrt{a+2x} + \sqrt{3x}} \right] \\&= \lim_{x \rightarrow a} \left[\frac{\sqrt{3a+x} + 2\sqrt{x}}{3(\sqrt{a+2x} + \sqrt{3x})} \right] \quad \dots \left[\begin{array}{l} \because x \rightarrow a, x \neq a \\ \therefore x-a \neq 0 \end{array} \right] \\&= \frac{\sqrt{3a+a} + 2\sqrt{a}}{3(\sqrt{a+2a} + \sqrt{3a})} \\&= \frac{\sqrt{4a} + 2\sqrt{a}}{3(\sqrt{3a} + \sqrt{3a})} \\&= \frac{2\sqrt{a} + 2\sqrt{a}}{3(2\sqrt{3a})} \\&= \frac{4\sqrt{a}}{6\sqrt{3}\sqrt{a}}\end{aligned}$$

$$= \frac{2}{3\sqrt{3}}.$$

Exercise 7.3 | Q 2.2 | Page 103

Evaluate the following limits: $\lim_{x \rightarrow 2} \left[\frac{x^2 - 4}{\sqrt{x+2} - \sqrt{3x-2}} \right]$

SOLUTION

$$\begin{aligned} & \lim_{x \rightarrow 2} \left[\frac{x^2 - 4}{\sqrt{x+2} - \sqrt{3x-2}} \right] \\ &= \lim_{x \rightarrow 2} \left[\frac{x^2 - 4}{\sqrt{x+2} - \sqrt{3x-2}} \times \frac{\sqrt{x+2} + \sqrt{3x-2}}{\sqrt{x+2} + \sqrt{3x-2}} \right] \\ &= \lim_{x \rightarrow 2} \frac{(x^2 - 4)(\sqrt{x+2} + \sqrt{3x-2})}{(x+2) - (3x-2)} \\ &= \lim_{x \rightarrow 2} \frac{(x^2 - 4)(\sqrt{x+2} + \sqrt{3x-2})}{-2x + 4} \\ &= \lim_{x \rightarrow 2} \frac{(x+2)(x-2)(\sqrt{x+2} + \sqrt{3x-2})}{-2(x-2)} \\ &= \lim_{x \rightarrow 2} \frac{(x+2)(\sqrt{x+2} + \sqrt{3x-2})}{-2} \quad \dots \left[\begin{array}{l} \text{As } x \rightarrow 2, x \neq 2 \\ \therefore x - 2 \neq 0 \end{array} \right] \\ &= \frac{(2+2)(\sqrt{2+2} + \sqrt{3(2)-2})}{-2} \\ &= \frac{4(2+2)}{-2} \\ &= \frac{16}{-2} \\ &= -8. \end{aligned}$$

Evaluate the following limits: $\lim_{x \rightarrow 1} \left[\frac{x^2 + x\sqrt{x} - 2}{x - 1} \right]$

SOLUTION

$$\begin{aligned}
 & \lim_{x \rightarrow 1} \left[\frac{x^2 + x\sqrt{x} - 2}{x - 1} \right] \\
 &= \lim_{x \rightarrow 1} \left[\frac{(x^2 - 1) + (x\sqrt{x} - 1)}{x - 1} \right] \\
 &= \lim_{x \rightarrow 1} \left[\frac{x^2 - 1}{x - 1} + \frac{x^{\frac{1}{2}} - 1}{x - 1} \right] \quad \dots \left[\because x\sqrt{x} = x^1 \cdot x^{\frac{1}{2}} = x^{1+\frac{1}{2}} = x^{\frac{3}{2}} \right] \\
 &= \lim_{x \rightarrow 1} \left(\frac{x^2 - 1^2}{x - 1} \right) + \lim_{x \rightarrow 1} \left(\frac{x^{\frac{3}{2}} - 1^{\frac{3}{2}}}{x - 1} \right) \\
 &= 2(1)1 + \frac{3}{2}(1)^{\frac{1}{2}} \quad \dots \left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right] \\
 &= 2 + \frac{3}{2} \\
 &= \frac{7}{2}.
 \end{aligned}$$

Evaluate the following limits: $\lim_{x \rightarrow 0} \left[\frac{\sqrt{1+x^2} - \sqrt{1+x}}{\sqrt{1+x^3} - \sqrt{1+x}} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \left[\frac{\sqrt{1+x^2} - \sqrt{1+x}}{\sqrt{1+x^3} - \sqrt{1+x}} \right] \\
&= \lim_{x \rightarrow 0} \frac{(\sqrt{1+x^2} - \sqrt{1+x})(\sqrt{1+x^3} + \sqrt{1+x})(\sqrt{1+x^2} + \sqrt{1+x})}{(\sqrt{1+x^3} - \sqrt{1+x})(\sqrt{1+x^3} + \sqrt{1+x})(\sqrt{1+x^2} + \sqrt{1+x})} \\
&= \lim_{x \rightarrow 0} \frac{[1+x^2 - (1+x)](\sqrt{1+x^3} + \sqrt{1+x})}{[1+x^3 - (1+x)](\sqrt{1+x^2} + \sqrt{1+x})} \\
&= \lim_{x \rightarrow 0} \frac{x(x-1)(\sqrt{1+x^3} + \sqrt{1+x})}{x(x^2-1)(\sqrt{1-x^2} + \sqrt{1+x})} \\
&= \lim_{x \rightarrow 0} \frac{\sqrt{1+x^3} + \sqrt{1+x}}{(x+1)(\sqrt{1+x^2} + \sqrt{1+x})} \quad \dots [\because x \rightarrow 0, \therefore x \neq 0, \therefore x-1 \neq 0] \\
&= \frac{\sqrt{1+0^3} + \sqrt{1+0}}{(0+1)(\sqrt{1+0^2} + \sqrt{1+0})} \\
&= \frac{1+1}{1(1+1)} \\
&= 1
\end{aligned}$$

Exercise 7.3 | Q 3.3 | Page 103

Evaluate the following limits: $\lim_{x \rightarrow 4} \left[\frac{x^2 + x - 20}{\sqrt{3x+4} - 4} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 4} \left[\frac{x^2 + x - 20}{\sqrt{3x + 4} - 4} \right] \\
&= \lim_{x \rightarrow 4} \left[\frac{(x + 5)(x - 4)}{\sqrt{3x + 4} - 4} \times \frac{\sqrt{3x + 4} + 4}{\sqrt{3x + 4} + 4} \right] \\
&= \lim_{x \rightarrow 4} \frac{(x + 5)(x - 4)(\sqrt{3x + 4} + 4)}{3x + 4 - 16} \\
&= \lim_{x \rightarrow 4} \frac{(x + 5)(x - 4)(\sqrt{3x + 4} + 4)}{3x - 12} \\
&= \lim_{x \rightarrow 4} \frac{(x + 5)(x - 4)(\sqrt{3x + 4} + 4)}{3(x - 4)} \\
&= \lim_{x \rightarrow 4} \frac{(x + 5)(\sqrt{3x + 4} + 4)}{3} \quad \dots \left[\begin{array}{l} \because x \rightarrow 4, x \neq 4 \\ \therefore x - 4 \neq 0 \end{array} \right] \\
&= \frac{(4 + 5)(\sqrt{3(4) + 4} + 4)}{3} \\
&= \frac{9(4 + 4)}{3} \\
&= 3(8) \\
&= 24.
\end{aligned}$$

Exercise 7.3 | Q 3.4 | Page 103

Evaluate the following limits: $\lim_{x \rightarrow 2} \left[\frac{x^3 - 8}{\sqrt{x + 2} - \sqrt{3x - 2}} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 2} \left[\frac{x^3 - 8}{\sqrt{x+2} - \sqrt{3x-2}} \right] \\
&= \lim_{x \rightarrow 2} \left[\frac{x^3 - 8}{\sqrt{x+2} - \sqrt{3x-2}} \times \frac{\sqrt{x+2} + \sqrt{3x-2}}{\sqrt{x+2} + \sqrt{3x-2}} \right] \\
&= \lim_{x \rightarrow 2} \frac{(x^3 - 8)(\sqrt{x+2} + \sqrt{3x-2})}{(x+2) - (3x-2)} \\
&= \lim_{x \rightarrow 2} \frac{(x^3 - 8)(\sqrt{x+2} + \sqrt{3x-2})}{-2x + 4} \\
&= \lim_{x \rightarrow 2} \frac{(x-2)(x^2 + 2x + 4)(\sqrt{x+2} + \sqrt{3x-2})}{-2(x-2)} \\
&= \lim_{x \rightarrow 2} \frac{(x^2 + 2x + 4)(\sqrt{x+2} + \sqrt{3x-2})}{-2} \quad \dots \left[\begin{array}{l} \text{As } x \rightarrow 2, x \neq 2 \\ \therefore x - 2 \neq 0 \end{array} \right] \\
&= -\frac{1}{2} \lim_{x \rightarrow 2} (x^2 + 2x + 4) \times \lim_{x \rightarrow 2} (\sqrt{x+2} + \sqrt{3x-2}) \\
&= -\frac{1}{2} \times [(2)^2 + 2(2) + 4] \times (\sqrt{2+2} + \sqrt{3(2)-2}) \\
&= -\frac{1}{2} \times 12 \times (2+2) \\
&= -6 \times 4 \\
&= -24.
\end{aligned}$$

Exercise 7.3 | Q 4.1 | Page 103

Evaluate the following limits: $\lim_{y \rightarrow 2} \left[\frac{2-y}{\sqrt{3-y}-1} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{y \rightarrow 2} \left[\frac{2-y}{\sqrt{3-y}-1} \right] \\
&= \lim_{y \rightarrow 2} \left[\frac{2-y}{\sqrt{3-y}-1} \times \frac{\sqrt{3-y}+1}{\sqrt{3-y}+1} \right] \\
&= \lim_{y \rightarrow 2} \frac{(2-y)(\sqrt{3-y}+1)}{3-y-1} \\
&= \lim_{y \rightarrow 2} \frac{(2-y)(\sqrt{3-y}+1)}{2-y} \\
&= \lim_{y \rightarrow 2} (\sqrt{3-y}+1) \quad \dots \left[\begin{array}{l} \text{As } y \rightarrow 2, y \neq 2 \\ \therefore y-2 \neq 0 \therefore 2-y \neq 0 \end{array} \right] \\
&= \sqrt{3-2}+1 \\
&= 1+1 \\
&= 2.
\end{aligned}$$

Exercise 7.3 | Q 4.2 | Page 103

Evaluate the following limits: $\lim_{z \rightarrow 4} \left[\frac{3 - \sqrt{5+z}}{1 - \sqrt{5-z}} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{z \rightarrow 4} \left[\frac{3 - \sqrt{5+z}}{1 - \sqrt{5-z}} \right] \\
&= \lim_{z \rightarrow 4} \left[\frac{3 - \sqrt{5+z}}{1 - \sqrt{5-z}} \times \frac{3 + \sqrt{5+z}}{3 + \sqrt{5+z}} \times \frac{1 + \sqrt{5-z}}{1 + \sqrt{5-z}} \right] \\
&= \lim_{z \rightarrow 4} \left[\frac{9 - (5+z)}{1 - (5-z)} \times \frac{1 + \sqrt{5-z}}{3 + \sqrt{5+z}} \right]
\end{aligned}$$

$$\begin{aligned}
&= \lim_{z \rightarrow 4} \left[\frac{4 - z}{-4 + z} \times \frac{1 + \sqrt{5 - z}}{3 + \sqrt{5 + z}} \right] \\
&= \lim_{z \rightarrow 4} \left[\frac{-(z - 4)}{z - 4} \times \frac{1 + \sqrt{5 - z}}{3 + \sqrt{5 + z}} \right] \\
&= \lim_{z \rightarrow 4} \left[\frac{-1 + \sqrt{5 - z}}{3 + \sqrt{5 + z}} \right] \quad \dots \left[\begin{array}{l} \because z \rightarrow 4, \therefore z \neq 4, \\ \therefore z - 4 \neq 0 \end{array} \right] \\
&= \frac{-(1 + \sqrt{5 - 4})}{3 + \sqrt{5 + 4}} \\
&= \frac{-(1 + 1)}{3 + 3} \\
&= \frac{-2}{6} \\
&= -\frac{1}{3}.
\end{aligned}$$

EXERCISE 7.4 [PAGE 105]

Exercise 7.4 | Q 1.1 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{9^x - 5^x}{4^x - 1} \right]$

SOLUTION

$$\begin{aligned}
&\lim_{x \rightarrow 0} \frac{9^x - 5^x}{4^x - 1} \\
&= \lim_{x \rightarrow 0} \frac{9^x - 1 + 1 - 5^x}{4^x - 1} \\
&= \lim_{x \rightarrow 0} \frac{(9^x - 1) - (5^x - 1)}{4^x - 1}
\end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{\frac{9^x - 1 - 5^x - 1}{x}}{\frac{4^x - 1}{x}} \quad \dots [\because x \rightarrow 0, \therefore x \neq 0] \\
&= \lim_{x \rightarrow 0} \frac{\left(\frac{9^x - 1}{x}\right) - \left(\frac{5^x - 1}{x}\right)}{\left(\frac{4^x - 1}{x}\right)} \\
&= \frac{\lim_{x \rightarrow 0} \frac{9^x - 1}{x} - \lim_{x \rightarrow 0} \frac{5^x - 1}{x}}{\lim_{x \rightarrow 0} \frac{4^x - 1}{x}} \\
&= \frac{\log 9 - \log 5}{\log 4} \quad \dots \left[\because \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \right] \\
&= \frac{1}{\log 4} \log \left(\frac{9}{5} \right).
\end{aligned}$$

Exercise 7.4 | Q 1.2 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{5^x + 3^x - 2^x - 1}{x} \right]$

SOLUTION

$$\begin{aligned}
&\lim_{x \rightarrow 0} \frac{5^x + 3^x - 2^x - 1}{x} \\
&\lim_{x \rightarrow 0} \frac{(5^x - 1) + (3^x - 2^x)}{x} \\
&= \lim_{x \rightarrow 0} \frac{(5^x - 1) + (3^x - 1) - (2^x - 1)}{x} \\
&= \lim_{x \rightarrow 0} \left(\frac{5^x - 1}{x} + \frac{3^x - 1}{x} - \frac{2^x - 1}{x} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{5^x - 1}{x} \right) + \lim_{x \rightarrow 0} \left(\frac{3^x - 1}{x} \right) - \lim_{x \rightarrow 0} \left(\frac{2^x - 1}{x} \right) \\
&= \log 5 + \log 3 - \log 2 \quad \dots \left[\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \right]
\end{aligned}$$

$$\begin{aligned}
 &= \log \frac{5 \times 3}{2} \\
 &= \log \frac{15}{2}.
 \end{aligned}$$

Exercise 7.4 | Q 1.3 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{\log(2+x) - \log(2-x)}{x} \right]$

SOLUTION

$$\begin{aligned}
 &\lim_{x \rightarrow 0} \left[\frac{\log(2+x) - \log(2-x)}{x} \right] \\
 &= \lim_{x \rightarrow 0} \frac{\log\left[2\left(1 + \frac{x}{2}\right)\right] - \log\left[2\left(1 - \frac{x}{2}\right)\right]}{x} \\
 &= \lim_{x \rightarrow 0} \frac{\log 2 + \log\left(1 + \frac{x}{2}\right) - [\log 2 + \log\left(1 - \frac{x}{2}\right)]}{x} \\
 &= \lim_{x \rightarrow 0} \frac{\log\left(1 + \frac{x}{2}\right) - \log\left(1 - \frac{x}{2}\right)}{x} \\
 &= \lim_{x \rightarrow 0} \left[\frac{\log\left(1 + \frac{x}{2}\right)}{x} - \frac{\log\left(1 - \frac{x}{2}\right)}{x} \right] \\
 &= \lim_{x \rightarrow 0} \left[\frac{\log\left(1 + \frac{x}{2}\right)}{2\left(\frac{x}{2}\right)} - \frac{\log\left(1 - \frac{x}{2}\right)}{(-2)\left(-\frac{x}{2}\right)} \right] \\
 &= \frac{1}{2} \lim_{x \rightarrow 0} \frac{\log\left(1 + \frac{x}{2}\right)}{\frac{x}{2}} + \frac{1}{2} \lim_{x \rightarrow 0} \frac{\log\left(1 - \frac{x}{2}\right)}{-\frac{x}{2}} \\
 &= \frac{1}{2}(1) + \frac{1}{2}(1) \dots \left[\because x \rightarrow 0, \frac{x}{2} \rightarrow 0 \text{ and } \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \right] \\
 &= 1.
 \end{aligned}$$

Evaluate the following: $\lim_{x \rightarrow 0} \frac{3^x + 3^{-x} - 2}{x^2}$

SOLUTION

$$\begin{aligned}
 & \lim_{x \rightarrow 0} \frac{3^x + 3^{-x} - 2}{x^2} \\
 &= \lim_{x \rightarrow 0} \frac{3^x + \frac{1}{3^x} - 2}{x^2} \\
 &= \lim_{x \rightarrow 0} \frac{(3^x)^2 + 1 - 2(3^x)}{3^x \cdot x^2} \\
 &= \lim_{x \rightarrow 0} \frac{(3^x - 1)^2}{x^2 \cdot (3^x)} \quad \dots [\because a^2 - 2ab + b^2 = (a - b)^2] \\
 &= \lim_{x \rightarrow 0} \left(\frac{3^x - 1}{x} \right)^2 \times \frac{1}{3^x} \\
 &= \lim_{x \rightarrow 0} \left(\frac{3^x - 1}{x} \right)^2 \times \frac{1}{\lim_{x \rightarrow 0} (3^x)} \\
 &= (\log 3)^2 \times \frac{1}{3^0} \\
 &= (\log 3)^2 \times \frac{1}{1} \quad \dots \left[\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \right] \\
 &= (\log 3)^2.
 \end{aligned}$$

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{3+x}{3-x} \right]^{\frac{1}{x}}$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \left(\frac{3+x}{3-x} \right)^{\frac{1}{x}} \\
&= \lim_{x \rightarrow 0} \left(\frac{1 + \frac{x}{3}}{1 - \frac{x}{3}} \right)^{\frac{1}{x}} \quad \dots \left[\begin{array}{c} \text{Divide numerator and} \\ \text{denominator by 3} \end{array} \right] \\
&= \lim_{x \rightarrow 0} \frac{\left(1 + \frac{x}{3} \right)^{\frac{1}{x}}}{\left(1 - \frac{x}{3} \right)^{\frac{1}{x}}} \\
&= \lim_{x \rightarrow 0} \frac{\left(1 + \frac{x}{3} \right)^{\frac{3}{x} \times \frac{1}{3}}}{\left(1 - \frac{x}{3} \right)^{\frac{-3}{x} \times \frac{1}{-3}}} \\
&= \frac{\lim_{x \rightarrow 0} \left[\left(1 + \frac{x}{3} \right)^{\frac{3}{x}} \right]^{\frac{1}{3}}}{\lim_{x \rightarrow 0} \left[\left(1 - \frac{x}{3} \right)^{\frac{-3}{x}} \right]^{-\frac{1}{3}}} \\
&= \frac{e^{\frac{1}{3}}}{e^{-\frac{1}{3}}} \quad \dots \left[\begin{array}{c} \because x \rightarrow 0, \frac{x}{3} \rightarrow 0, \frac{-x}{3} \rightarrow 0 \text{ and} \\ \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e \end{array} \right] \\
&= e^{\frac{1}{3} + \frac{1}{3}} \\
&= e^{\frac{2}{3}}.
\end{aligned}$$

Exercise 7.4 | Q 2.3 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{\log(3-x) - \log(3+x)}{x} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \frac{\log(3-x) - \log(3+x)}{x} \\
&= \lim_{x \rightarrow 0} \frac{1}{x} \log \left(\frac{3-x}{3+x} \right)
\end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \log \left(\frac{3-x}{3+x} \right)^{\frac{1}{x}} \\
&= \lim_{x \rightarrow 0} \log \left(\frac{1 - \frac{x}{3}}{1 + \frac{x}{3}} \right)^{\frac{1}{x}} \\
&= \log \left[\lim_{x \rightarrow 0} \frac{\left(1 - \frac{x}{3}\right)^{\frac{1}{x}}}{\left(1 + \frac{x}{3}\right)^{\frac{1}{x}}} \right] \\
&= \log \left[\frac{\left\{ \lim_{x \rightarrow 0} \left(1 - \frac{x}{3}\right)^{\frac{-3}{x}} \right\}^{\frac{-1}{3}}}{\left\{ \lim_{x \rightarrow 0} \left(1 + \frac{x}{3}\right)^{\frac{3}{x}} \right\}^{\frac{1}{3}}} \right] \\
&= \log \left(\frac{e^{-\frac{1}{3}}}{e^{\frac{1}{3}}} \right) \dots \left[\because x \rightarrow 0, \pm \frac{x}{3} \rightarrow 0 \text{ and } \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e \right] \\
&= \log e^{-\frac{2}{3}} \\
&= -\frac{2}{3} \cdot \log e \\
&= -\frac{2}{3} (1) \\
&= -\frac{2}{3}.
\end{aligned}$$

Exercise 7.4 | Q 3.1 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{a^{3x} - b^{2x}}{\log 1 + 4x} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \left[\frac{a^{3x} - b^{2x}}{\log 1 + 4x} \right] \\
&= \lim_{x \rightarrow 0} \frac{a^{3x} - 1 - b^{2x} + 1}{\log 1 + 4x} \\
&= \lim_{x \rightarrow 0} \frac{\frac{a^{3x} - 1}{x} - \frac{b^{2x} - 1}{x}}{\frac{\log 1 + 4x}{x}} \\
&= \frac{\lim_{x \rightarrow 0} \left[\frac{a^{3x} - 1}{x} - \frac{b^{2x} - 1}{x} \right]}{\lim_{x \rightarrow 0} \frac{\log 1 + 4x}{4x}} \\
&= \frac{\lim_{x \rightarrow 0} \left[\frac{a^{3x} - 1}{3x} \right] \times 3 - \lim_{x \rightarrow 0} \left[\frac{b^{2x} - 1}{2x} \right] \times 2}{\lim_{x \rightarrow 0} \frac{\log 1 + 4x}{4x} \times 4} \\
&= \frac{3 \log a - 2 \log b}{1 \times 4} \dots \left[\begin{array}{l} \because x \rightarrow 0, 2x \rightarrow 0, 3x \rightarrow 0 \\ 4x \rightarrow 0 \text{ and } \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \\ \text{and } \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \end{array} \right] \\
&= \frac{1}{4} (\log a^3 - \log b^2) \\
&= \frac{1}{4} \log \left(\frac{a^3}{b^2} \right).
\end{aligned}$$

Exercise 7.4 | Q 3.2 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{(2^x - 1)^2}{(3^x - 1) \times \log(1 + x)} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \left[\frac{(2^x - 1)^2}{(3^x - 1) \times \log(1 + x)} \right] \\
&= \lim_{x \rightarrow 0} \frac{\frac{(2^x - 1)^2}{x^2}}{\frac{3^x - 1 \cdot \log 1 + x}{x^2}} \cdots \left[\begin{array}{l} \text{Divide Numerator and} \\ \text{Denominator by } x^2 \\ \text{As } x \rightarrow 0, x \neq 0 \\ \therefore x^2 \neq 0 \end{array} \right] \\
&= \frac{\lim_{x \rightarrow 0} \left(\frac{2^x - 1}{x} \right)^2}{\lim_{x \rightarrow 0} \left[\left(\frac{3^x - 1}{x} \right) \times \frac{\log 1 + x}{x} \right]} \\
&= \frac{\lim_{x \rightarrow 0} \left(\frac{2^x - 1}{x} \right)^2}{\lim_{x \rightarrow 0} \left(\frac{3^x - 1}{x} \right) \times \lim_{x \rightarrow 0} \frac{\log 1 + x}{x}} \\
&= \frac{(\log 2)^2}{\log 3 \times 1} \cdots \left[\begin{array}{l} \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a, \\ \lim_{x \rightarrow 0} \frac{\log(1 + x)}{x} = 1 \end{array} \right] \\
&= \frac{(\log 2)^2}{\log 3}.
\end{aligned}$$

Exercise 7.4 | Q 3.3 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{15^x - 5^x - 3^x + 1}{x^2} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \left[\frac{15^x - 5^x - 3^x + 1}{x^2} \right] \\
&= \lim_{x \rightarrow 0} \frac{5^x \cdot 3^x - 5^x - 3^x + 1}{x^2} \\
&= \lim_{x \rightarrow 0} \frac{5^x(3^x - 1) - 1(3^x - 1)}{x^2} \\
&= \lim_{x \rightarrow 0} \frac{(3^x - 1)(5^x - 1)}{x^2} \\
&= \lim_{x \rightarrow 0} \left(\frac{3^x - 1}{x} \times \frac{5^x - 1}{x} \right) \\
&= \lim_{x \rightarrow 0} \frac{3^x - 1}{x} \times \lim_{x \rightarrow 0} \frac{5^x - 1}{x} \\
&= \log 3 \cdot \log 5 \dots \left[\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \right]
\end{aligned}$$

Exercise 7.4 | Q 3.4 | Page 105

Evaluate the following: $\lim_{x \rightarrow 2} \left[\frac{3^{\frac{x}{2}} - 3}{3^x - 9} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 2} \left[\frac{3^{\frac{x}{2}} - 3}{3^x - 9} \right] \\
&= \lim_{x \rightarrow 2} \left[\frac{3^{\frac{x}{2}} - 3}{(3^{\frac{x}{2}})^2 - (3)^2} \right] \\
&= \lim_{x \rightarrow 2} \frac{3^{\frac{x}{2}} - 3}{(3^{\frac{x}{2}} - 3)(3^{\frac{x}{2}} + 3)} \\
&= \lim_{x \rightarrow 2} \frac{1}{3^{\frac{x}{2}} + 3} \dots \left[\begin{array}{l} \text{As } x \rightarrow 2, \frac{x}{2} \rightarrow 1 \\ \therefore 3^{\frac{x}{2}} \rightarrow 3^1 \therefore 3^{\frac{x}{2}} \neq 3 \\ \therefore 3^{\frac{x}{2}} - 3 \neq 0 \end{array} \right]
\end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{3^{\frac{2}{2}} + 3} \\
 &= \frac{1}{3^1 + 3} \\
 &= \frac{1}{6}.
 \end{aligned}$$

Exercise 7.4 | Q 4.1 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{(25)^x - 2(5)^x + 1}{x^2} \right]$

SOLUTION

$$\begin{aligned}
 &\lim_{x \rightarrow 0} \left[\frac{(25)^x - 2(5)^x + 1}{x^2} \right] \\
 &= \lim_{x \rightarrow 0} \left[\frac{(5)^{2x} - 2(5)^x + 1}{x^2} \right] \\
 &= \lim_{x \rightarrow 0} \left[\frac{(5^x)^2 - 2(5)^x + 1}{x^2} \right] \\
 &= \lim_{x \rightarrow 0} \frac{(5^x - 1)^2}{x^2} \\
 &= \lim_{x \rightarrow 0} \left(\frac{5^x - 1}{x} \right)^2 \\
 &= \log 5^2 \quad \dots \left[\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \right]
 \end{aligned}$$

Exercise 7.4 | Q 4.2 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{(49)^x - 2(35)^x + (25)^x}{x^2} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \left[\frac{(49)^x - 2(35)^x + (25)^x}{x^2} \right] \\
&= \lim_{x \rightarrow 0} \left[\frac{(7^2)^x - 2(7 \times 5)^x + (5^2)^x}{x^2} \right] \\
&= \lim_{x \rightarrow 0} \left[\frac{(7^x)^2 - 2(7^x - 5^x)^x + (5^x)^2}{x^2} \right] \\
&= \lim_{x \rightarrow 0} \frac{(7^x - 5^x)^2}{x^2} \\
&= \lim_{x \rightarrow 0} \left[\frac{7^x - 1 - 5^x + 1}{x} \right]^2 \\
&= \lim_{x \rightarrow 0} \left[\frac{7^x - 1}{x} - \frac{5^x - 1}{x} \right]^2 \\
&= \left[\lim_{x \rightarrow 0} \frac{7^x - 1}{x} - \lim_{x \rightarrow 0} \frac{5^x - 1}{x} \right]^2 \\
&= (\log 7 - \log 5)^2 \quad \dots \left[\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \right] \\
&= \left(\log \frac{7}{5} \right)^2.
\end{aligned}$$

MISCELLANEOUS EXERCISE 7 [PAGES 105 - 106]

Exercise 7.4 | Q 1.1 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{9^x - 5^x}{4^x - 1} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \frac{9^x - 5^x}{4^x - 1} \\
&= \lim_{x \rightarrow 0} \frac{9x - 1 + 1 - 5x}{4^x - 1} \\
&= \lim_{x \rightarrow 0} \frac{(9^x - 1) - (5^x - 1)}{4^x - 1} \\
&= \lim_{x \rightarrow 0} \frac{\frac{9^x - 1}{x} - \frac{5^x - 1}{x}}{\frac{4^x - 1}{x}} \quad \dots [\because x \rightarrow 0, \therefore x \neq 0] \\
&= \lim_{x \rightarrow 0} \frac{\left(\frac{9^x - 1}{x}\right) - \left(\frac{5^x - 1}{x}\right)}{\left(\frac{4^x - 1}{x}\right)} \\
&= \frac{\lim_{x \rightarrow 0} \frac{9^x - 1}{x} - \lim_{x \rightarrow 0} \frac{5^x - 1}{x}}{\lim_{x \rightarrow 0} \frac{4^x - 1}{x}} \\
&= \frac{\log 9 - \log 5}{\log 4} \quad \dots \left[\because \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \right] \\
&= \frac{1}{\log 4} \log \left(\frac{9}{5} \right).
\end{aligned}$$

Exercise 7.4 | Q 1.2 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{5^x + 3^x - 2^x - 1}{x} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \frac{5^x + 3^x - 2^x - 1}{x} \\
&= \lim_{x \rightarrow 0} \frac{(5^x - 1) + (3^x - 2^x)}{x} \\
&= \lim_{x \rightarrow 0} \frac{(5^x - 1) + (3^x - 1) - (2^x - 1)}{x}
\end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \left(\frac{5^x - 1}{x} + \frac{3^x - 1}{x} - \frac{2^x - 1}{x} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{5^x - 1}{x} \right) + \lim_{x \rightarrow 0} \left(\frac{3^x - 1}{x} \right) - \lim_{x \rightarrow 0} \left(\frac{2^x - 1}{x} \right) \\
&= \log 5 + \log 3 - \log 2 \quad \dots \left[\lim_{x \rightarrow 0} \frac{a^x - 1}{x} - \log a \right] \\
&= \log \frac{5 \times 3}{2} \\
&= \log \frac{15}{2}.
\end{aligned}$$

Exercise 7.4 | Q 1.3 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{\log(2+x) - \log(2-x)}{x} \right]$

SOLUTION

$$\begin{aligned}
&\lim_{x \rightarrow 0} \left[\frac{\log(2+x) - \log(2-x)}{x} \right] \\
&= \lim_{x \rightarrow 0} \frac{\log \left[2 \left(1 + \frac{x}{2} \right) \right] - \log \left[2 \left(1 - \frac{x}{2} \right) \right]}{x} \\
&= \lim_{x \rightarrow 0} \frac{\log 2 + \log \left(1 + \frac{x}{2} \right) - [\log 2 + \log \left(1 - \frac{x}{2} \right)]}{x} \\
&= \lim_{x \rightarrow 0} \frac{\log \left(1 + \frac{x}{2} \right) - \log \left(1 - \frac{x}{2} \right)}{x} \\
&= \lim_{x \rightarrow 0} \left[\frac{\log \left(1 + \frac{x}{2} \right)}{x} - \frac{\log \left(1 - \frac{x}{2} \right)}{x} \right] \\
&= \lim_{x \rightarrow 0} \left[\frac{\log \left(1 + \frac{x}{2} \right)}{2 \left(\frac{x}{2} \right)} - \frac{\log \left(1 - \frac{x}{2} \right)}{(-2) \left(-\frac{x}{2} \right)} \right] \\
&= \frac{1}{2} \lim_{x \rightarrow 0} \frac{\log \left(1 + \frac{x}{2} \right)}{\frac{x}{2}} + \frac{1}{2} \lim_{x \rightarrow 0} \frac{\log \left(1 - \frac{x}{2} \right)}{-\frac{x}{2}}
\end{aligned}$$

$$= \frac{1}{2}(1) + \frac{1}{2}(1) \dots \left[\begin{array}{l} \because x \rightarrow 0, \frac{x}{2} \rightarrow 0 \text{ and} \\ \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \end{array} \right]$$

$$= 1.$$

Exercise 7.4 | Q 2.1 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \frac{3^x + 3^{-x} - 2}{x^2}$

SOLUTION

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{3^x + 3^{-x} - 2}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{3^x + \frac{1}{3^x} - 2}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{(3^x)^2 + 1 - 2(3^x)}{3^x \cdot x^2} \\ &= \lim_{x \rightarrow 0} \frac{(3^x - 1)^2}{x^2 \cdot (3^x)} \quad \dots [\because a^2 - 2ab + b^2 = (a - b)^2] \\ &= \lim_{x \rightarrow 0} \left(\frac{3^x - 1}{x} \right)^2 \times \frac{1}{3^x} \\ &= \lim_{x \rightarrow 0} \left(\frac{3^x - 1}{x} \right)^2 \times \frac{1}{\lim_{x \rightarrow 0} (3^x)} \\ &= (\log 3)^2 \times \frac{1}{3^0} \\ &= (\log 3)^2 \times \frac{1}{1} \dots \left[\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \right] \\ &= (\log 3)^2. \end{aligned}$$

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{3+x}{3-x} \right]^{\frac{1}{x}}$

SOLUTION

$$\begin{aligned}
 & \lim_{x \rightarrow 0} \left(\frac{3+x}{3-x} \right)^{\frac{1}{x}} \\
 &= \lim_{x \rightarrow 0} \left(\frac{1 + \frac{x}{3}}{1 - \frac{x}{3}} \right)^{\frac{1}{x}} \quad \dots \left[\begin{array}{l} \text{Divide numerator and} \\ \text{denominator by 3} \end{array} \right] \\
 &= \lim_{x \rightarrow 0} \frac{\left(1 + \frac{x}{3} \right)^{\frac{1}{x}}}{\left(1 - \frac{x}{3} \right)^{\frac{1}{x}}} \\
 &= \lim_{x \rightarrow 0} \frac{\left(1 + \frac{x}{3} \right)^{\frac{3}{x} \times \frac{1}{3}}}{\left(1 - \frac{x}{3} \right)^{\frac{-3}{x} \times \frac{1}{-3}}} \\
 &= \frac{\lim_{x \rightarrow 0} \left[\left(1 + \frac{x}{3} \right)^{\frac{3}{x}} \right]^{\frac{1}{3}}}{\lim_{x \rightarrow 0} \left[\left(1 - \frac{x}{3} \right)^{\frac{-3}{x}} \right]^{-\frac{1}{3}}} \\
 &= \frac{e^{\frac{1}{3}}}{e^{\frac{-1}{3}}} \quad \dots \left[\begin{array}{l} \because x \rightarrow 0, \frac{x}{3} \rightarrow 0, \frac{-x}{3} \rightarrow 0 \text{ and } \\ \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e \end{array} \right] \\
 &= e^{\frac{1}{3} + \frac{1}{3}} \\
 &= e^{\frac{2}{3}}.
 \end{aligned}$$

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{\log(3-x) - \log(3+x)}{x} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \frac{\log(3-x) - \log(3+x)}{x} \\
&= \lim_{x \rightarrow 0} \frac{1}{x} \log\left(\frac{3-x}{3+x}\right) \\
&= \lim_{x \rightarrow 0} \log\left(\frac{3-x}{3+x}\right)^{\frac{1}{x}} \\
&= \lim_{x \rightarrow 0} \log\left(\frac{1 - \frac{x}{3}}{1 + \frac{x}{3}}\right)^{\frac{1}{x}} \\
&= \log \left[\lim_{x \rightarrow 0} \frac{\left(1 - \frac{x}{3}\right)^{\frac{1}{x}}}{\left(1 + \frac{x}{3}\right)^{\frac{1}{x}}} \right] \\
&= \log \left[\frac{\left\{ \lim_{x \rightarrow 0} \left(1 - \frac{x}{3}\right)^{\frac{-3}{x}} \right\}^{\frac{-1}{3}}}{\left\{ \lim_{x \rightarrow 0} \left(1 + \frac{x}{3}\right)^{\frac{3}{x}} \right\}^{\frac{1}{3}}} \right] \\
&= \log \left(\frac{e^{-\frac{1}{3}}}{e^{\frac{1}{3}}} \right) \dots \left[\because x \rightarrow 0, \pm \frac{x}{3} \rightarrow 0 \text{ and } \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e \right] \\
&= \log e^{\frac{-2}{3}} \\
&= -\frac{2}{3} \cdot \log e \\
&= -\frac{2}{3} (1) \\
&= -\frac{2}{3}.
\end{aligned}$$

Exercise 7.4 | Q 3.1 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{a^{3x} - b^{2x}}{\log 1 + 4x} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \left[\frac{a^{3x} - b^{2x}}{\log 1 + 4x} \right] \\
&= \lim_{x \rightarrow 0} \frac{a^{3x} - 1 - b^{2x} + 1}{\log 1 + 4x} \\
&= \lim_{x \rightarrow 0} \frac{\frac{a^{3x}-1-b^{2x}+1}{x}}{\frac{\log 1+4x}{x}} \\
&= \frac{\lim_{x \rightarrow 0} \left[\frac{a^{3x}-1}{x} - \frac{b^{2x}-1}{x} \right]}{\lim_{x \rightarrow 0} \frac{\log 1+4x}{4x}} \\
&= \frac{\lim_{x \rightarrow 0} \left[\frac{a^{3x}-1}{3x} \right] \times 3 - \lim_{x \rightarrow 0} \left[\frac{b^{2x}-1}{2x} \right] \times 2}{\lim_{x \rightarrow 0} \frac{\log 1+4x}{4x} \times 4} \\
&= \frac{3 \log a - 2 \log b}{1 \times 4} \dots \left[\begin{array}{l} \because x \rightarrow 0, 2x \rightarrow 0, 3x \rightarrow 0 \\ 4x \rightarrow 0 \text{ and } \lim_{x \rightarrow 0} \frac{a^x-1}{x} = \log a \\ \text{and } \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \end{array} \right] \\
&= \frac{1}{4} (\log a^3 - \log b^2) \\
&= \frac{1}{4} \log \left(\frac{a^3}{b^2} \right).
\end{aligned}$$

Exercise 7.4 | Q 3.2 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{(2^x - 1)^2}{(3^x - 1) \times \log(1 + x)} \right]$

SOLUTION

$$\lim_{x \rightarrow 0} \left[\frac{(2^x - 1)^2}{(3^x - 1) \times \log(1 + x)} \right]$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{\frac{(2^x-1)^2}{x^2}}{\frac{3^x-1 \cdot \log 1+x}{x^2}} \cdots \left[\begin{array}{l} \text{Divide Numerator and} \\ \text{Denominator by } x^2 \\ \text{As } x \rightarrow 0, x \neq 0 \\ \therefore x^2 \neq 0 \end{array} \right] \\
&= \frac{\lim_{x \rightarrow 0} \left(\frac{2^x-1}{x} \right)^2}{\lim_{x \rightarrow 0} \left[\left(\frac{3^x-1}{x} \right) \times \frac{\log 1+x}{x} \right]} \\
&= \frac{\lim_{x \rightarrow 0} \left(\frac{2^x-1}{x} \right)^2}{\lim_{x \rightarrow 0} \left(\frac{3^x-1}{x} \right) \times \lim_{x \rightarrow 0} \frac{\log 1+x}{x}} \\
&= \frac{(\log 2)^2}{\log 3 \times 1} \cdots \left[\begin{array}{l} \lim_{x \rightarrow 0} \frac{a^x-1}{x} = \log a, \\ \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \end{array} \right] \\
&= \frac{(\log 2)^2}{\log 3}.
\end{aligned}$$

Exercise 7.4 | Q 3.3 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{15^x - 5^x - 3^x + 1}{x^2} \right]$

SOLUTION

$$\begin{aligned}
&\lim_{x \rightarrow 0} \left[\frac{15^x - 5^x - 3^x + 1}{x^2} \right] \\
&= \lim_{x \rightarrow 0} \frac{5^x \cdot 3^x - 5^x - 3^x + 1}{x^2} \\
&= \lim_{x \rightarrow 0} \frac{5^x(3^x - 1) - 1(3^x - 1)}{x^2} \\
&= \lim_{x \rightarrow 0} \frac{(3^x - 1)(5^x - 1)}{x^2}
\end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \left(\frac{3^x - 1}{x} \times \frac{5^x - 1}{x} \right) \\
&= \lim_{x \rightarrow 0} \frac{3^x - 1}{x} \times \lim_{x \rightarrow 0} \frac{5^x - 1}{x} \\
&= \log 3 \cdot \log 5 \dots \left[\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \right]
\end{aligned}$$

Exercise 7.4 | Q 3.4 | Page 105

Evaluate the following: $\lim_{x \rightarrow 2} \left[\frac{3^{\frac{x}{2}} - 3}{3^x - 9} \right]$

SOLUTION

$$\begin{aligned}
&\lim_{x \rightarrow 2} \left[\frac{3^{\frac{x}{2}} - 3}{3^x - 9} \right] \\
&= \lim_{x \rightarrow 2} \left[\frac{3^{\frac{x}{2}} - 3}{(3^{\frac{x}{2}})^2 - (3)^2} \right] \\
&= \lim_{x \rightarrow 2} \frac{3^{\frac{x}{2}} - 3}{(3^{\frac{x}{2}} - 3)(3^{\frac{x}{2}} + 3)} \\
&= \lim_{x \rightarrow 2} \frac{1}{3^{\frac{x}{2}} + 3} \dots \left[\begin{array}{l} \text{As } x \rightarrow 2, \frac{x}{2} \rightarrow 1 \\ \therefore 3^{\frac{x}{2}} \rightarrow 3^1 \therefore 3^{\frac{x}{2}} \neq 3 \\ \therefore 3^{\frac{x}{2}} - 3 \neq 0 \end{array} \right] \\
&= \frac{1}{3^{\frac{2}{2}} + 3} \\
&= \frac{1}{3^1 + 3} \\
&= \frac{1}{6}.
\end{aligned}$$

Exercise 7.4 | Q 4.1 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{(25)^x - 2(5)^x + 1}{x^2} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \left[\frac{(25)^x - 2(5)^x + 1}{x^2} \right] \\
&= \lim_{x \rightarrow 0} \left[\frac{(5)^{2x} - 2(5)^x + 1}{x^2} \right] \\
&= \lim_{x \rightarrow 0} \left[\frac{(5^x)^2 - 2(5)^x + 1}{x^2} \right] \\
&= \lim_{x \rightarrow 0} \frac{(5^x - 1)^2}{x^2} \\
&= \lim_{x \rightarrow 0} \left(\frac{5^x - 1}{x} \right)^2 \\
&= \log 5^2 \quad \dots \left[\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \right]
\end{aligned}$$

Exercise 7.4 | Q 4.2 | Page 105

Evaluate the following: $\lim_{x \rightarrow 0} \left[\frac{(49)^x - 2(35)^x + (25)^x}{x^2} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \left[\frac{(49)^x - 2(35)^x + (25)^x}{x^2} \right] \\
&= \lim_{x \rightarrow 0} \left[\frac{(7^2)^x - 2(7 \times 5)^x + (5^2)^x}{x^2} \right] \\
&= \lim_{x \rightarrow 0} \left[\frac{(7^x)^2 - 2(7^x - 5^x)^x + (5^x)^2}{x^2} \right] \\
&= \lim_{x \rightarrow 0} \frac{(7^x - 5^x)^2}{x^2}
\end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \left[\frac{7^x - 1 - 5^x - 1}{x} \right]^2 \\
&= \lim_{x \rightarrow 0} \left[\frac{7^x - 1}{x} - \frac{5^x - 1}{x} \right]^2 \\
&= \left[\lim_{x \rightarrow 0} \frac{7^x - 1}{x} - \lim_{x \rightarrow 0} \frac{5^x - 1}{x} \right]^2 \\
&= (\log 7 - \log 5)^2 \dots \left[\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \right] \\
&= \left(\log \frac{7}{5} \right)^2.
\end{aligned}$$

MISCELLANEOUS EXERCISE 7 [PAGES 105 - 106]

Miscellaneous Exercise 7 | Q 1 | Page 105

if $\lim_{x \rightarrow 2} \frac{x^n - 2}{x - 2} = 80$ then find the value of n.

SOLUTION

$$\begin{aligned}
&\lim_{x \rightarrow 2} \frac{x^n - 2}{x - 2} = 80 \\
&\therefore n(2)^{n-1} = 80 \quad \dots \left[\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right] \\
&\therefore n(2)^{n-1} = 5 \times 16 \\
&= 5 \times (2)^4 \\
&\therefore n(2)^{n-1} = 5 \times (2)^{5-1} \\
&\therefore n = 5.
\end{aligned}$$

Miscellaneous Exercise 7 | Q 2.01 | Page 105

Evaluate the following Limits: $\lim_{x \rightarrow a} \frac{(x+2)^{\frac{5}{3}} - (a+2)^{\frac{5}{3}}}{x - a}$

SOLUTION

$$\lim_{x \rightarrow a} \frac{(x+2)^{\frac{5}{3}} - (a+2)^{\frac{5}{3}}}{x-a}$$

Put $x+2 = y$ and $a+2 = b$

As $x \rightarrow a$, $x+2 \rightarrow a+2$

i.e. $y \rightarrow b$.

$$\begin{aligned} \therefore \lim_{x \rightarrow a} \frac{(x+2)^{\frac{5}{3}} - (a+2)^{\frac{5}{3}}}{x-a} \\ &= \lim_{x \rightarrow a} \frac{y^{\frac{5}{3}} - b^{\frac{5}{3}}}{(y-2) - (b-2)} \\ &= \lim_{y \rightarrow b} \frac{y^{\frac{5}{3}} - b^{\frac{5}{3}}}{y-b} \\ &= \frac{5}{3} b^{\frac{2}{3}} \dots \left[\lim_{x \rightarrow a} \frac{x^n - a^n}{x-a} = na^{n-1} \right] \\ &= \frac{5}{3} (a+2)^{\frac{2}{3}} \dots [\because b = a+2] \end{aligned}$$

Miscellaneous Exercise 7 | Q 2.02 | Page 106

Evaluate the following Limits: $\lim_{x \rightarrow 2} \frac{(1+x)^n - 1}{x}$

SOLUTION

$$\lim_{x \rightarrow 2} \frac{(1+x)^n - 1}{x}$$

Put $1+x = y$ $\therefore x = y-1$

As $x \rightarrow 2$, $y \rightarrow 1$

$$\therefore \lim_{x \rightarrow 2} \frac{(1+x)^n - 1}{x}$$

$$\begin{aligned}
&= \lim_{y \rightarrow 1} \frac{y^n - 1}{y - 1} \\
&= \lim_{y \rightarrow 1} \frac{y^n - 1^n}{y - 1} \\
&= n(1)^{n-1} \quad \dots \left[\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right] \\
&= n.
\end{aligned}$$

Miscellaneous Exercise 7 | Q 2.03 | Page 106

Evaluate the following Limits: $\lim_{x \rightarrow 2} \left[\frac{(x - 2)}{2x^2 - 7x + 6} \right]$

SOLUTION

$$\begin{aligned}
&\lim_{x \rightarrow 2} \left[\frac{(x - 2)}{2x^2 - 7x + 6} \right] \\
&= \lim_{x \rightarrow 2} \frac{(x - 2)}{(x - 2)(2x - 3)} \\
&= \lim_{x \rightarrow 2} \frac{1}{2x - 3} \quad \dots \left[\begin{array}{l} \text{As } x \rightarrow 2, x \neq 2 \\ \therefore x - 2 \neq 0 \end{array} \right] \\
&= \frac{1}{2(2) - 3} \\
&= 1.
\end{aligned}$$

Miscellaneous Exercise 7 | Q 2.04 | Page 106

Evaluate the following Limits: $\lim_{x \rightarrow 1} \left[\frac{x^3 - 1}{x^2 + 5x - 6} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 1} \left[\frac{x^3 - 1}{x^2 + 5x - 6} \right] \\
&= \lim_{x \rightarrow 1} \frac{(x - 1)(x^2 + x + 1)}{(x - 1)(x + 6)} \\
&= \lim_{x \rightarrow 1} \frac{x^2 + x + 1}{x + 6} \quad \dots \left[\begin{array}{l} \text{As } x \rightarrow 1, x \neq 1 \\ \therefore x - 1 \neq 0 \end{array} \right] \\
&= \frac{(1)^2 + 1 + 1}{1 + 6} \\
&= \frac{3}{7}.
\end{aligned}$$

Miscellaneous Exercise 7 | Q 2.05 | Page 106

Evaluate the following Limits: $\lim_{x \rightarrow 3} \left[\frac{x - 3}{\sqrt{x - 2} - \sqrt{4 - x}} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 3} \left[\frac{x - 3}{\sqrt{x - 2} - \sqrt{4 - x}} \right] \\
&= \lim_{x \rightarrow 3} \left[\frac{x - 3}{\sqrt{x - 2} - \sqrt{4 - x}} \times \frac{\sqrt{x - 2} + \sqrt{4 - x}}{\sqrt{x - 2} + \sqrt{4 - x}} \right] \\
&= \lim_{x \rightarrow 3} \frac{(x - 3)(\sqrt{x - 2} + \sqrt{4 - x})}{(x - 2) - (4 - x)} \\
&= \lim_{x \rightarrow 3} \frac{(x - 3)(\sqrt{x - 2} + \sqrt{4 - x})}{2x - 6} \\
&= \lim_{x \rightarrow 3} \frac{(x - 3)(\sqrt{x - 2} + \sqrt{4 - x})}{2(x - 3)}
\end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 3} \frac{\sqrt{x-2} + \sqrt{4-x}}{2} \dots \left[\begin{array}{l} \text{As } x \rightarrow 3, x \neq 3 \\ \therefore x-3 \neq 0 \end{array} \right] \\
&= \frac{1}{2} \lim_{x \rightarrow 3} (\sqrt{x-2} + \sqrt{4-x}) \\
&= \frac{1}{2} (\sqrt{3-2} + \sqrt{4-3}) \\
&= \frac{1}{2} (1 + 1) \\
&= 1.
\end{aligned}$$

Miscellaneous Exercise 7 | Q 2.06 | Page 106

Evaluate the following Limits: $\lim_{x \rightarrow 4} \left[\frac{3 - \sqrt{5+x}}{1 - \sqrt{5-x}} \right]$

SOLUTION

$$\begin{aligned}
&\lim_{x \rightarrow 4} \left[\frac{3 - \sqrt{5+x}}{1 - \sqrt{5-x}} \right] \\
&= \lim_{x \rightarrow 4} \left[\frac{3 - \sqrt{5+x}}{1 - \sqrt{5-x}} \times \frac{3 + \sqrt{5+x}}{3 + \sqrt{5+x}} \times \frac{1 + \sqrt{5-x}}{1 + \sqrt{5-x}} \right] \\
&= \lim_{x \rightarrow 4} \left[\frac{9 - (5+x)}{1 - (5-x)} \times \frac{1 + \sqrt{5-x}}{3 + \sqrt{5+x}} \right] \\
&= \lim_{x \rightarrow 4} \left[\frac{4-x}{-4+x} \times \frac{1 + \sqrt{5-x}}{3 + \sqrt{5+x}} \right] \\
&= \lim_{x \rightarrow 4} \left[\frac{-(x-4)}{x-4} \times \frac{1 + \sqrt{5-x}}{3 + \sqrt{5+x}} \right]
\end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 4} \left[\frac{-(1 + \sqrt{5 - x})}{3 + \sqrt{5 + x}} \right] \dots \left[\begin{array}{l} \text{As } x \rightarrow 4, x \neq 4 \\ \therefore x - 4 \neq 0 \end{array} \right] \\
&= \frac{-(1 + \sqrt{5 - 4})}{3 + \sqrt{5 + 4}} \\
&= \frac{-(1 + \sqrt{5 - 4})}{3 + \sqrt{5 + 4}} \\
&= \frac{-(1 + 1)}{3 + 3} \\
&= \frac{-2}{6} \\
&= -\frac{1}{3}.
\end{aligned}$$

Miscellaneous Exercise 7 | Q 2.07 | Page 106

Evaluate the following Limits: $\lim_{x \rightarrow 0} \left[\frac{5^x - 1}{x} \right]$

SOLUTION

$$\begin{aligned}
&\lim_{x \rightarrow 0} \left[\frac{5^x - 1}{x} \right] \\
&= \log 5 \quad \dots \left[\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \right]
\end{aligned}$$

Miscellaneous Exercise 7 | Q 2.08 | Page 106

Evaluate the following Limits: $\lim_{x \rightarrow 0} \left(1 + \frac{x}{5} \right)^{\frac{1}{x}}$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \left(1 + \frac{x}{5}\right)^{\frac{1}{x}} \\
&= \lim_{x \rightarrow 0} \left[\left(1 + \frac{x}{5}\right)^{\frac{5}{x}} \right]^{\frac{1}{5}} \\
&= e^{\frac{1}{5}} \quad \dots \left[\lim_{x \rightarrow 0} \left(1 + \frac{x}{5}\right)^{\frac{5}{x}} = e \right]
\end{aligned}$$

Miscellaneous Exercise 7 | Q 2.09 | Page 106

Evaluate the following Limits: $\lim_{x \rightarrow 0} \left[\frac{\log(1 + 9x)}{x} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \left[\frac{\log(1 + 9x)}{x} \right] \\
&= \lim_{x \rightarrow 0} \left[\frac{\log(1 + 9x)}{9x} \right] \times 9 \\
&= 1 \times 9 \quad \dots \left[\lim_{x \rightarrow 0} \frac{\log(1 + x)}{x} = 1 \right] \\
&= 9.
\end{aligned}$$

Miscellaneous Exercise 7 | Q 2.1 | Page 106

Evaluate the following Limits: $\lim_{x \rightarrow 0} \frac{(1 - x)^5 - 1}{(1 - x)^3 - 1}$

SOLUTION

$$\lim_{x \rightarrow 0} \left[\frac{(1 - x)^5 - 1}{(1 - x)^3 - 1} \right]$$

Put $1 - x = y$

As $x \rightarrow 0, y \rightarrow 1$

$$\begin{aligned}
& \therefore \lim_{x \rightarrow 0} \left[\frac{(1-x)^5 - 1}{(1-x)^3 - 1} \right] \\
&= \lim_{y \rightarrow 1} \frac{y^5 - 1}{y^3 - 1} \\
&= \lim_{y \rightarrow 1} \left(\frac{\frac{y^5-1}{y-1}}{\frac{y^3-1}{y-1}} \right) \cdots \left[\begin{array}{l} \text{As } y \rightarrow 1, y \neq 1 \\ \therefore y - 1 \neq 0 \\ \text{Divide Numerator and} \\ \text{Denominator by } y - 1 \end{array} \right] \\
&= \frac{\lim_{y \rightarrow 1} \frac{y^5-1}{y-1}}{\lim_{y \rightarrow 1} \frac{y^3-1}{y-1}} \\
&= \frac{5(1)^4}{3(1)^2} \cdots \left[\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right] \\
&= \frac{5}{3}.
\end{aligned}$$

Miscellaneous Exercise 7 | Q 2.11 | Page 106

Evaluate the following Limits: $\lim_{x \rightarrow 0} \left[\frac{a^x + b^x + c^x - 3}{x} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \left[\frac{a^x + b^x + c^x - 3}{x} \right] \\
&= \lim_{x \rightarrow 0} \frac{(a^x - 1) + (b^x - 1) + (c^x - 1)}{x} \\
&= \lim_{x \rightarrow 0} \left(\frac{a^x - 1}{x} + \frac{b^x - 1}{x} + \frac{c^x - 1}{x} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{a^x - 1}{x} \right) + \lim_{x \rightarrow 0} \left(\frac{b^x - 1}{x} \right) + \lim_{x \rightarrow 0} \left(\frac{c^x - 1}{x} \right)
\end{aligned}$$

$$= \log a + \log b + \log c \quad \dots \left[\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \right]$$

$$= \log (abc).$$

Miscellaneous Exercise 7 | Q 2.12 | Page 106

Evaluate the following Limits: $\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2}{x^2}$

SOLUTION

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{e^x + \frac{1}{e^x} - 2}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{(e^x)^2 + 1 - 2e^x}{x^2 \cdot e^x} \\ &= \lim_{x \rightarrow 0} \frac{(e^x - 1)^2}{x^2 \cdot e^x} \\ &= \lim_{x \rightarrow 0} \left[\left(\frac{e^x - 1}{x} \right)^2 \times \frac{1}{e^x} \right] \\ &= \lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} \right)^2 \times \frac{1}{\lim_{x \rightarrow 0} e^x} \\ &= (1)^2 \times \frac{1}{e^0} \quad \dots \left[\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \right] \\ &= 1 \times \frac{1}{1} \\ &= 1. \end{aligned}$$

Miscellaneous Exercise 7 | Q 2.13 | Page 106

Evaluate the following Limits: $\lim_{x \rightarrow 0} \left[\frac{x(6^x - 3^x)}{(2^x - 1) \cdot \log(1 + x)} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \frac{x(6^x - 3^x)}{(2^x - 1) \cdot \log(1 + x)} \\
&= \lim_{x \rightarrow 0} \frac{x(3^x \cdot 2^x - 3^x)}{(2^x - 1) \cdot \log(1 + x)} \\
&= \lim_{x \rightarrow 0} \frac{x \cdot 3^x(2^x - 1)}{(2^x - 1) \cdot \log(1 + x)} \\
&= \lim_{x \rightarrow 0} \frac{x \cdot 3^x}{\log(1 + x)} \quad \dots \left[\begin{array}{l} \text{As } x \rightarrow 0, 2^x \rightarrow 2^0 \\ \text{i.e. } 2^x \rightarrow 1 \therefore 2^x \neq 1 \\ \therefore 2^x - 1 \neq 0 \end{array} \right] \\
&= \lim_{x \rightarrow 0} \frac{3^x}{\frac{\log(1+x)}{x}} \\
&= \frac{\lim_{x \rightarrow 0} 3^x}{\lim_{x \rightarrow 0} \frac{\log(1+x)}{x}} \\
&= \frac{3^0}{1} \quad \dots \left[\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \right] \\
&= 1.
\end{aligned}$$

Miscellaneous Exercise 7 | Q 2.14 | Page 106

Evaluate the following Limits: $\lim_{x \rightarrow 0} \left[\frac{a^{3x} - a^{2x} - a^x + 1}{x^2} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \frac{a^{3x} - a^{2x} - a^x + 1}{x^2} \\
&= \lim_{x \rightarrow 0} \frac{a^{2x} \cdot a^x - a^{2x} - a^x + 1}{x^2}
\end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{a^{2x}(a^x - 1) - 1(a^x - 1)}{x^2} \\
&= \lim_{x \rightarrow 0} \frac{(a^x - 1) \cdot (a^{2x} - 1)}{x^2} \\
&= \lim_{x \rightarrow 0} \left(\frac{a^x - 1}{x} \times \frac{a^{2x} - 1}{x} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{a^x - 1}{x} \right) \times \lim_{x \rightarrow 0} \left(\frac{a^{2x} - 1}{2x} \right) \times 2 \\
&= \log a \cdot (2 \log a) \quad \dots \left[\begin{array}{l} \text{As } x \rightarrow 0, 2x \rightarrow 0 \text{ and} \\ \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \end{array} \right] \\
&= 2(\log a)^2.
\end{aligned}$$

Miscellaneous Exercise 7 | Q 2.15 | Page 106

Evaluate the following Limits: $\lim_{x \rightarrow 0} \left[\frac{(5^x - 1)^2}{x \cdot \log(1 + x)} \right]$

SOLUTION

$$\begin{aligned}
&\lim_{x \rightarrow 0} \frac{(5^x - 1)^2}{x \cdot \log(1 + x)} \\
&= \lim_{x \rightarrow 0} \frac{\frac{(5^x - 1)^2}{x^2}}{\frac{x \cdot \log(1 + x)}{x^2}} \quad \dots \left[\begin{array}{l} \text{As } x \rightarrow 0, x \neq 0 \therefore x^2 \neq 0 \\ \text{Divide Numerator and} \\ \text{Denominator by } x^2 \end{array} \right] \\
&= \frac{\lim_{x \rightarrow 0} \left(\frac{5^x - 1}{x} \right)^2}{\lim_{x \rightarrow 0} \frac{\log(1 + x)}{x}}
\end{aligned}$$

$$= \frac{(\log 5)^2}{1} \cdots \left[\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a, \right]$$

$$= (\log 5)^2.$$

Miscellaneous Exercise 7 | Q 2.16 | Page 106

Evaluate the following Limits: $\lim_{x \rightarrow 0} \left[\frac{a^{4x} - 1}{b^{2x} - 1} \right]$

SOLUTION

$$\lim_{x \rightarrow 0} \frac{a^{4x} - 1}{b^{2x} - 1}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{a^{4x} - 1}{x}}{\frac{b^{2x} - 1}{x}}$$

$$= \frac{\lim_{x \rightarrow 0} \left(\frac{a^{4x} - 1}{4x} \right) \times 4}{\lim_{x \rightarrow 0} \left(\frac{b^{2x} - 1}{2x} \right) \times 2}$$

$$= \frac{4 \log a}{2 \log b} \cdots \left[\begin{array}{l} \text{As } x \rightarrow 0, 2x \rightarrow 0, 4x \rightarrow 0 \\ \text{and } \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \end{array} \right]$$

$$= \frac{2 \log a}{\log b}.$$

Miscellaneous Exercise 7 | Q 2.17 | Page 106

Evaluate the following Limits: $\lim_{x \rightarrow 0} \left[\frac{\log 100 + \log(0.01 + x)}{x} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \left[\frac{\log 100 + \log(0.01 + x)}{x} \right] \\
&= \lim_{x \rightarrow 0} \frac{\log[100(0.001 + x)]}{x} \\
&= \lim_{x \rightarrow 0} \frac{\log(1 + 100x)}{x} \\
&= \lim_{x \rightarrow 0} \left[\frac{\log(1 + 100x)}{100x} \right] \times 100 \\
&= 1 \times 100 \quad \dots \left[\begin{array}{l} \text{As } x \rightarrow 0, 100x \rightarrow 0 \text{ and} \\ \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \end{array} \right] \\
&= 100.
\end{aligned}$$

Miscellaneous Exercise 7 | Q 2.18 | Page 106

Evaluate the following Limits: $\lim_{x \rightarrow 0} \left[\frac{\log(4 - x) - \log(4 + x)}{x} \right]$

SOLUTION

$$\begin{aligned}
& \lim_{x \rightarrow 0} \frac{\log(4 - x) - \log(4 + x)}{x} \\
&= \lim_{x \rightarrow 0} \frac{\log\left[4\left(1 - \frac{x}{4}\right)\right] - \log\left[4\left(1 + \frac{x}{4}\right)\right]}{x} \\
&= \lim_{x \rightarrow 0} \frac{\log 4 + \log\left(1 - \frac{x}{4}\right) - [\log 4 + \log\left(1 + \frac{x}{4}\right)]}{x} \\
&= \lim_{x \rightarrow 0} \frac{\log\left(1 - \frac{x}{4}\right) - \log\left(1 + \frac{x}{4}\right)}{x} \\
&= \lim_{x \rightarrow 0} \left[\frac{\log\left(1 - \frac{x}{4}\right)}{x} - \frac{\log\left(1 + \frac{x}{4}\right)}{x} \right]
\end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{\log\left(1 - \frac{x}{4}\right)}{(-4)\left(-\frac{x}{4}\right)} - \lim_{x \rightarrow 0} \frac{\log\left(1 + \frac{x}{4}\right)}{4\left(\frac{x}{4}\right)} \\
&= -\frac{1}{4} \lim_{x \rightarrow 0} \frac{\log\left(1 - \frac{x}{4}\right)}{-\frac{x}{4}} - \frac{1}{4} \lim_{x \rightarrow 0} \frac{\log\left(1 + \frac{x}{4}\right)}{\frac{x}{4}} \\
&= -\frac{1}{4}(1) - \frac{1}{4}(1) \quad \dots \left[\begin{array}{l} \text{As } x \rightarrow 0, \frac{x}{4} \rightarrow 0, \frac{-x}{4} \rightarrow 0 \\ \text{and } \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \end{array} \right] \\
&= -\frac{1}{2}.
\end{aligned}$$

Miscellaneous Exercise 7 | Q 2.19 | Page 106

Evaluate the limit of the function if exist at $x = 1$ where, $f(x) = \begin{cases} 7 - 4x & x < 1 \\ x^2 + 2 & x \geq 1 \end{cases}$

SOLUTION

$$f(x) = 7 - 4x : x < 1$$

$$= x^2 + 2 : x \geq 1$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (7 - 4x)$$

$$= 7 - 4(1)$$

$$= 3$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (x^2 + 2)$$

$$= (1)^2 + 2$$

$$= 3$$

$$\therefore \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x)$$

$$\therefore \lim_{x \rightarrow 1} f(x) \text{ exists.}$$

$$\therefore \lim_{x \rightarrow 1} f(x) = 3.$$